

Devarsh Vora

devarshvora45@gmail.com | 682-406-7329 | Indiana, USA (Open to relocate) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

Data Scientist with 3+ years of experience connecting customer behavior, product experiments, and healthcare payment data to operational decisions. Develops Python and SQL solutions for churn prediction, collections forecasting, and NLP, evaluating models against meaningful baselines and business constraints. Combines causal inference and A/B testing with production monitoring to distinguish measurable improvements from statistical noise. Works with business and engineering teams to determine which changes merit rollout, which accounts need intervention, and when models need refinement.

Professional Experience

Data Scientist, CyberdomeUSA

08/2025 – Present | Indiana, USA

- Developed daily dbt transformations in SQL for 120,000 monthly product events, validating account joins and metric definitions to supply activation funnels and retention reporting for weekly product reviews.
- Ran account-randomized A/B tests for onboarding changes, defining sample sizes and primary activation metrics, checking assignment balance, and presenting confidence intervals and support-volume guardrails to guide product release decisions.
- Deployed XGBoost churn scores into CRM outreach queues, achieving 0.80 AUROC versus a 0.71 logistic regression baseline on temporal holdouts, with SHAP explanations helping account managers prioritize retention outreach.
- Refreshed behavioral account segments monthly using K-means, testing cluster stability and tracking adoption by segment in Tableau so customer-success managers could tailor outreach and review subsequent changes in engagement.
- Monitored deployed churn models for feature drift and calibration as renewal outcomes matured, tracking versions in MLflow and investigating segment-level degradation to inform retraining decisions and changes to outreach thresholds.

Applied AI Analyst (Contract), Matrices

09/2025 – 03/2026 | Remote, SF, USA

- Applied stratified sampling and inter-annotator agreement analysis to client support conversations, maintaining versioned intent datasets and resolving ambiguous labels with operations specialists to support consistent training and evaluation.
- Trained scikit-learn intent classifiers on sentence embeddings, increasing held-out macro-F1 by 0.09 over TF-IDF logistic regression, then deployed confidence-based routing with specialist review and feedback for subsequent retraining.
- Evaluated client workflow launches using difference-in-differences and matched cohorts, checking pre-intervention trends and workload mix and reporting adjusted handling-time estimates with confidence intervals to guide rollout and staffing decisions.
- Used paired comparisons and human scoring rubrics to evaluate LLM-generated summaries on versioned test sets, identifying factual errors across client segments and recommending prompt revisions before production releases.

Graduate Assistant, The University of Texas at Arlington

03/2024 – 07/2025 | Texas, USA

- Maintained Python and SQL preparation scripts for recurring academic datasets, validating identifiers and missing values before publishing versioned extracts and data dictionaries used by project teams for scheduled analyses.
- Supported study analyses in R using regression and bootstrap confidence intervals, checking model assumptions and documenting sampling limitations so academic teams could interpret findings consistently across reporting cycles.
- Published Tableau reports on participation and submission trends, standardizing cohort definitions and reconciling totals with source records so coordinators could identify missing submissions before monthly reporting deadlines.

Data Scientist, CrystalVox Limited

12/2020 – 12/2022 | Gujarat, India

- Deployed daily LightGBM denial-risk scores into billing review queues, improving temporal-holdout precision-recall AUC from 0.24 to 0.36 over logistic regression using submission-time features and flagging high-risk claims before filing.
- Investigated payer-specific denial spikes with SQL cohort analysis, separating eligibility and coding errors, then worked with billing leads to update submission checks and monitor recurring issues through weekly reviews.
- Published weekly SARIMAX collections forecasts for finance teams, reducing rolling-backtest MAE by 16% versus seasonal-naive forecasts, reporting prediction intervals, and reviewing forecast errors as payment batches closed each month.
- Maintained provider entity-resolution logic using normalized identifiers and fuzzy matching, validating ambiguous pairs through billing review and applying approved mappings to claims reports with source-record traceability for investigations.
- Delivered Power BI dashboards for denial rates, receivables aging, and payer turnaround, reconciling SQL measures with billing totals and highlighting overdue balances for client account reviews and collection prioritization.

Technical Skills

- **Programming & Analysis:** Python, SQL, R, Bash, pandas, NumPy, SciPy, statsmodels, Exploratory Data Analysis
- **Statistics & Experimentation:** A/B Testing, Hypothesis Testing, Causal Inference, Bayesian Inference, Regression, Power Analysis, Bootstrapping, Confidence Intervals, Cohort Analysis
- **Machine Learning & AI:** scikit-learn, XGBoost, LightGBM, PyTorch, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Time Series Forecasting, Anomaly Detection, SHAP, NLP, Embeddings, RAG, Prompt Engineering, LLM Evaluation
- **Data & Visualization:** PostgreSQL, Snowflake, PySpark, dbt, ETL/ELT, Data Validation, Tableau, Power BI, Matplotlib, Seaborn
- **Cloud & MLOps:** AWS, Docker, MLflow, Git, CI/CD, Model Deployment, Batch Inference, Model Monitoring, Drift Detection

Education

Master of Science, University of Texas at Arlington

08/2023 - 05/2025 | Texas, USA

Computer Science | Specialty: Data Science

Bachelor of Technology, Gujarat Technological University

07/2019 - 07/2023 | Gujarat, India

Information & Communication Technology | Specialty: AI/ML